Fiscal Year:	FY 2011	**Task Last Updated:**	FY 05/19/2011
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Novel Explorations Into the Interactions Between Light and Gravity Sensing in Plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
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City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Space Life Sciences-Flight NNH09ZTT002N
Start Date:	08/01/2010	**End Date:**	08/27/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Wigley, Cecilia	**Contact Phone:**	650-604-3019
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Original end date was 7/31/2013; changed to 8/27/2012 per PI's move to U Mississippi and grant transfer there (Ed., 9/28/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University)		
Grant/Contract No.:	NNX10AM86G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this project are to determine how gravity and light responses influence each other in plants and to better understand the cellular signaling mechanisms involved in plant tropisms. This proposed project builds on our previous spaceflight experiments on the ISS with the European Modular Cultivation System (EMCS) using the model plant Arabidopsis In this new proposed project, we plan to confirm and extend our discovery during the recent ISS experiment of a novel red-light-based positive phototropic response in plant shoots. Our hypothesis is that positive red-light-sensing, which was known in older plant lineages, is masked by normal 1-g conditions in more recently evolved lineages. We also propose to investigate fundamental interactions among red and blue light signaling pathways and the gravity sensing mechanisms. The ISS is the only laboratory facility where it is possible to reliably and simultaneously vary light and gravity stimuli for the proposed experiments. These experiments can be accomplished with telemetric science and therefore fit with the extremely limited down mass capabilities of current spaceflight opportunities. The EMCS is an automated facility and can be operated largely with telemetry. Experimental containers can be launched via Progress, ATV, or HTV spacecraft, and during an experimental run, images of seedling growth responses can be downlinked to Earth. A significant advantage of this proposed project is that we will use existing flight hardware that has been flown successfully and allowed for growth of Arabidopsis seedlings in the EMCS. This project is also relevant to the recent emphasis at NASA to study plant growth and development at fractional g-levels such as those found on the moon and Mars. Improved knowledge of the basic mechanistic processes that will be the focus of this project is vital to develop ways to use plants in extraterrestrial bioregenerative life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We plan to study the effects of gravity and light on basic cellular mechanisms in plants, so this project is relevant to the emphasis on using Arabidopsis thaliana as a model system in developmental biology. Since plants will be a necessary part of bioregenerative life support needed to send humans to Mars and beyond, the knowledge obtained from our space flight experiments will be critical for developing ways to effectively use plants in the bioregenerative life support systems. These life support systems also can be used in ground-based Earth applications such as the design of large-scale greenhouses to support crop plants and agriculture. In addition, improved knowledge of light and gravity systems in plants will, in the long term, allow us to develop better crop plants for use on Earth.
Task Progress:	This spaceflight project has been given the code name: SEEDLING GROWTH. In this past year, we have defined the operational requirements and protocols for the spaceflight experiments as summarized below. Operational Requirements: • Deliver 16 ECs/EUE/Seed Cassettes to ISS on SpaceX3. • For launch, ECs should be oriented such that launch loads are perpendicular to the baseplate. • At TBD times, images will be automatically downlinked. • Functional EMCS required to perform experiment runs. • Functional MELFI required to store frozen samples. • GLACIER required for return of frozen samples to Earth. Operational Protocols: • Destow EC/EUE/ Assembly • Install ECs in EMCS Facility • At conclusion of automated experimental sequence, remove ECs from EMCS Facility • Remove Seed Cassettes from ECs and freeze in MELFI • Stow ECs in ambient stowage • For return to Earth, place frozen Seed Cassettes in GLACIER • For return to Earth, place ECs in ambient stowage
Bibliography Type:	Description: (Last Updated: 11/25/2025)
Abstracts for Journals and Proceedings	Huntoon KE, Millar KDL, Kiss JZ. "Phototropism is enhanced in seedlings grown in fractional gravity on the ISS." American Society of Plant Biologists, 2011 Midwestern Section Annual Meeting, W. Lafayette, IN, March 19-20, 2011. Abstract Book. American Society of Plant Biologists 2011 Midwestern Section Annual Meeting, W. Lafayette, IN, March 19-20, 2011. , Mar-2011
Abstracts for Journals and Proceedings	Hopkins JA, Kiss JZ. "Tissue specific altered phytochrome pathways impact root phototropism in Arabidopsis thaliana." American Society of Plant Biologists 2011 Midwestern Section Annual Meeting, W. Lafayette, IN, March 19-20, 2011. Abstract Book. American Society of Plant Biologists 2011 Midwestern Section Annual Meeting, W. Lafayette, IN, March 19-20, 2011. , Mar-2011
Abstracts for Journals and Proceedings	Pyle T, Millar KDL, Kumar P, Kiss JZ, Souret F, Correll MJ. "The effects of gravity and the spaceflight environment on the gene expression profile of Arabidopsis thaliana seedlings." 26th Annual Meeting of the American Society for Gravitational and Space Biology, Washington, DC, November 4-7, 2010. Program and Abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, 2010. p. 35. , Nov-2011

Articles in Peer-reviewed Journals	Kiss JZ, Millar KD, Kumar P, Edelmann RE, Correll MJ. "Improvements in the re-flight of spaceflight experiments on plant tropisms." Advances in Space Research 2011 Feb 1;47(3):545-52. http://dx.doi.org/10.1016/j.asr.2010.09.024 , Feb-2011
Articles in Peer-reviewed Journals	Kumar P, Millar KDL, Kiss JZ. "Inflorescence stems of the mdr1 mutant display altered gravitropism and phototropism." Environmental and Experimental Botany 2011 Feb;70(2-3):244–50. http://dx.doi.org/10.1016/j.envexpbot.2010.09.019 , Feb-2011
Articles in Peer-reviewed Journals	Millar KD, Kumar P, Correll MJ, Mullen JL, Hangarter RP, Edelmann RE, Kiss JZ. "A novel phototropic response to red light is revealed in microgravity." New Phytol. 2010 May;186(3):648-56. PMID: 20298479 , May-2010
Articles in Peer-reviewed Journals	Matía I, González-Camacho F, Herranz R, Kiss JZ, Gasset G, van Loon J Marco R, Medina FJ. "Plant cell proliferation and growth are altered by microgravity conditions in spaceflight." Journal of Plant Physiology 2010 Oct;167(3):184–93. http://dx.doi.org/10.1016/j.jplph.2009.08.012 , Oct-2010